The Hawaiian Earring VS a Bouquet with Countably Many Petals

Visually very similar, mathematically very different

Lenin David Angarita Rivera / National University of Colombia

Master's in Mathematical Sciences
Ecogyt - 2026
Ldangaritar@unal.edu.co

Motivation

CW complexes constitute one of the most important classes of spaces in algebraic topology, since a large part of the tools of homotopy theory are developed on them. In this context, the **CW Approximation Theorem** states that every topological space admits an approximation by a CW complex.
This construction raises a natural question: given a space that is not a CW complex, how can a CW approximation be described, and which topological properties does it preserve?
A paradigmatic example is the **Hawaiian earring**, a space that, despite its simple geometric appearance, exhibits highly nontrivial homotopical and homological phenomena. Its similarity to the bouquet with countably many petals naturally leads us to ask:

Can the bouquet with countably many petals be a CW approximation of the Hawaiian earring?

The **aim** of this poster is to conduct a bibliographic survey that allows us to answer this question.

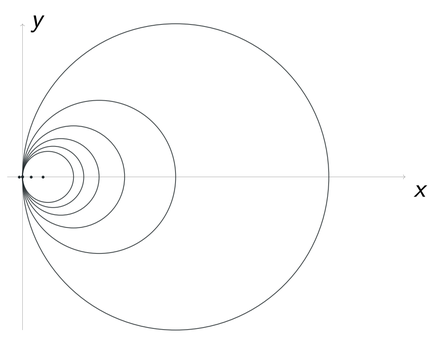
Introduction

Consider the following topological spaces.

Hawaiian earring. It is defined by

$$\mathbb{H} = \bigcup_{n=1}^{\infty} \left\{ (x,y) \in \mathbb{R}^2 : \left(x - \frac{1}{n}\right)^2 + y^2 = \left(\frac{1}{n}\right)^2 \right\},$$

endowed with the subspace topology inherited from $\mathbb{R}^2$.

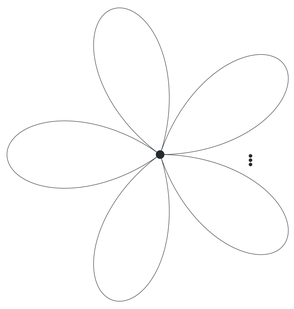


Hawaiian earring.

Bouquet with countably many petals. It is defined by

$$\bigvee_{i=1}^{\infty} \mathbb{S}^1,$$

endowed with the quotient topology.



Bouquet with countably many petals.

Are these spaces homeomorphic? Are they homotopy equivalent? Is there a weak homotopy equivalence between them?

Preliminaries

The following fundamental results will be used in this poster.

Theorem (CW Approximation Theorem)
For every topological space X, there exist a CW complex Y and a map

$$f : Y \longrightarrow X$$

that induces isomorphisms

$$f_* : \pi_n(Y) \xrightarrow{\cong} \pi_n(X), \qquad n \geq 0.$$

Definition
A continuous map

$$f : X \longrightarrow Y$$

is called a **weak homotopy equivalence** if the induced map

$$f_* : \pi_0(X) \longrightarrow \pi_0(Y)$$

is a bijection and, for every basepoint $x_0 \in X$ and every $n \geq 1$, it induces an isomorphism

$$f_* : \pi_n(X, x_0) \xrightarrow{\cong} \pi_n\big(Y, f(x_0)\big).$$

Theorem (Homology invariance)
A weak homotopy equivalence $f : X \longrightarrow Y$ induces isomorphisms

$$f_* : H_n(X; G) \xrightarrow{\cong} H_n(Y; G)$$

for every n and every coefficient group G.

Theorem
Let X be a CW complex. For every point $x \in X$, there exist arbitrarily small contractible neighborhoods of x; therefore, X is locally contractible.

Topological level: homeomorphism

Proposition
The Hawaiian earring and the bouquet with countably many petals are not homeomorphic.

$$\mathbb{H} \ncong \bigvee_{i=1}^{\infty} \mathbb{S}^1$$

The Hawaiian earring $\mathbb{H}$ is compact, since it is a closed and bounded subset of $\mathbb{R}^2$. In the bouquet, choose one point x_i on each petal. The sequence $(x_i)_{i\geq 1}$ has no convergent subsequence.

Homotopy level: fundamental group

The spaces are not homotopy equivalent either:

$$\pi_1\left(\bigvee_{i=1}^{\infty} \mathbb{S}^1\right) \ncong \pi_1(\mathbb{H}, (0,0))$$

- The group $\pi_1(\mathbb{H}, (0,0))$ is **not free** [1].
- For the bouquet with countably many petals, $\pi_1\left(\bigvee_{i=1}^{\infty} \mathbb{S}^1\right) \cong *_{i=1}^{\infty} \mathbb{Z}$.

The intuitive reason is that in $\mathbb{H}$ there are loops that traverse *infinitely many* circles (each one smaller than the previous one), and they are not homotopic to any finite product of generators, something impossible in a free group, where every element is a finite word in the generators.

Homological level

The first homology group of the Hawaiian earring is highly nontrivial:

$$H_1(\mathbb{H}) \cong \prod_{i=1}^{\infty} \mathbb{Z} \oplus \prod_{p:\text{prime}} A_p \oplus \bigoplus_{2^{\aleph_0}} \mathbb{Q}$$

Here, each A_p is the p-adic completion of $\bigoplus_{2^{\aleph_0}} \mathbb{J}_p$ [2].
In contrast,

$$H_1\left(\bigvee_{i=1}^{\infty} \mathbb{S}^1\right) \cong \bigoplus_{i=1}^{\infty} \mathbb{Z}$$

CW approximation and weak equivalence

The Hawaiian earring $\mathbb{H}$ is not a CW complex, since it is not locally contractible at the accumulation point.
The bouquet with countably many petals does admit a CW-complex structure.
The difference between their homology groups shows that the bouquet cannot be a CW approximation of the Hawaiian earring.
In particular, these spaces are not weakly equivalent either.

Conclusions

Invariant	Hawaiian earring	Bouquet with countably many petals
Basic topology	Compact	Not compact
Fundamental group	$\pi_1(\mathbb{H})$ is not free	$*_{i=1}^{\infty} \mathbb{Z}$
Singular homology	$H_1(\mathbb{H})$ contains product-type terms	$\bigoplus_{i=1}^{\infty} \mathbb{Z}$
CW complex	No	Yes
CW approximation	The bouquet is not a CW approximation	It does not weakly approximate $\mathbb{H}$

Questions and directions for further study

The content presented in this poster constitutes a first stage in the study of the Hawaiian earring. The work currently continues in the following directions.

- **Fundamental group.** Further investigate the connection between the group of transfinite words and $\pi_1(\mathbb{H})$ described by Cannon and Conner [1].
- **Direct limit problem.** This semester, spaces such as the complement of a *wild knot* were studied. They exhibit this accumulation phenomenon but still admit a description as a direct limit. This leads to the following question:

  What property of the Hawaiian earring prevents its fundamental group from being obtained as a direct limit of free groups?

- **First homology group.** Following Cannon and Conner, we view $\pi_1(\mathbb{H})$ as a group of transfinite words [1]. The description of $H_1(\mathbb{H})$ by Eda and Kawamura contains a nonstandard summand [2].

  How can this summand be identified in terms of abelianized transfinite words?

References

[1] J. W. Cannon and G. R. Conner, The combinatorial structure of the hawaiian earring group, *Topology and its Applications*, vol. 106, no. 3, pp. 225–271, 2000.

[2] K. Eda and K. Kawamura, The singular homology of the hawaiian earring, *Journal of the London Mathematical Society*, vol. 62, no. 1, pp. 305–310, 2000.

[3] A. Hatcher, *Algebraic Topology*. Cambridge University Press, 2002.

[4] K. Eda, The fundamental groups of one-dimensional wild spaces and the hawaiian earring, *Topology and its Applications*, vol. 123, no. 3, pp. 479–505, 2002.